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Case study: a hybrid plant in Hat Yai whose good output was hidden by a wrong capacity setting

18 October 2026 · Rev02 · Prepared by Solar ESS from remote monitoring data. Customer details removed.

Summary

The plant is working well. It is registered in its monitoring app as 10 kWp, but it has a single PV string whose data points to about 6 kWp, so the app makes it look as if it produces only 63% of what it should. Measured against about 6 kWp, it yields about 4.0 kWh per kWp per day, which is normal to good for Hat Yai. Since a battery was added on 18 May 2026, the shop has covered 98% of its own electricity.

Two items need attention first:

1. **Export was blocked for 24 days (28 Mar – 20 Apr 2026).** The inverter held production down to the shop's own use, losing about 500 kWh.
2. **The registered capacity is wrong.** Correcting it from 10 kWp to the installed size makes the app's performance figures meaningful.

Plant at a glance

Review of all monitoring data from commissioning on 29 Jan 2026 to 17 Oct 2026 (262 days), from SolisCloud.

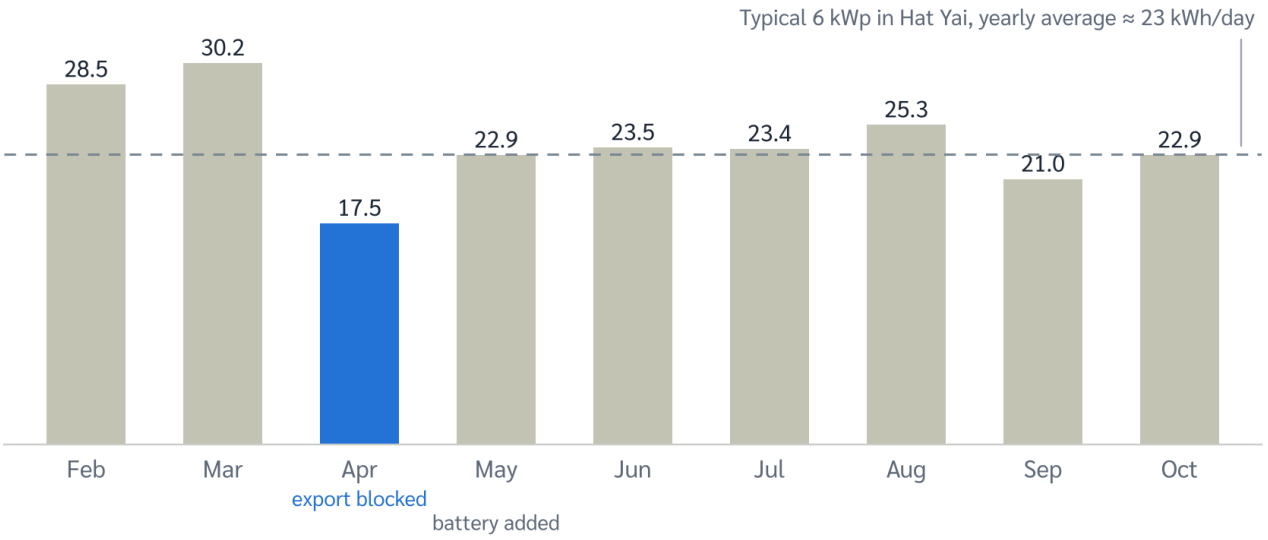
Item	Value
Location	Hat Yai, Songkhla
PV array	One string on MPPT1, about 6 kWp (estimated from string data); registered as 10 kWp
Inverter	Solis S6-EH3P10K hybrid, 10 kW, three-phase; MPPT2 unused
Battery	One 48 V battery, added 18 May 2026
Total production	6,230 kWh (about 4.0 kWh per kWp per day at 6 kWp)
Best day	36 kWh (25 Mar 2026), about 6 kWh per kWp
Total consumption	6,183 kWh (about 22–23 kWh/day)
Grid import / export	1,402 kWh / 1,755 kWh
Self-sufficiency since battery	98% (import 0.5 kWh/day, was 12.4)
Active alarms	None

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Monthly performance

Production matches a typical 6 kWp plant, except in April

Average 23.9 kWh/day, Feb–Oct 2026, against about 23 kWh/day for a typical 6 kWp plant · unit: kWh per day



Source: SolisCloud daily totals, Feb–Oct 2026 (Oct = 1–17 Oct). The typical value is an approximate yearly average.

Production is at or above a typical 6 kWp plant in most months, and well above it in the dry season. April is the exception, because export was blocked for most of the month.

Findings

1. The registered capacity is too high

The plant has one PV string, connected to input 1. At midday on sunny October days its output levels off at 4.6–5.3 kW, with 14–16 A at about 360 V, which fits a string of about 6 kWp. The best day so far, 36 kWh, is about 6 kWh per kWp at that size. Divided by the registered 10 kWp, the same output looks like a 37% shortfall.

2. Export was blocked for 24 days

From 28 Mar to 20 Apr 2026 the plant exported nothing, and each day's production equalled the shop's daytime use. Production fell to 11 kWh/day, against 30–34 kWh/day in the two weeks before and after. That is about 500 kWh lost. The inverter's operation log in SolisCloud is empty, so the setting was changed on the inverter itself or from the installer's app.

3. Selling back to the grid

The plant has exported 1,755 kWh since January. Without a PEA sell-back contract, this energy earns nothing.

4. Battery energy counters do not add up

Since May the battery shows 1,832 kWh discharged but only 1,526 kWh charged. A battery cannot give out more than it takes in, so its readings are wrong. SolisCloud lists the battery maker as "OTHER". If the inverter misreads the battery's charge level, the battery can be over-discharged or never fully charged.

5. Steady overnight load

The shop draws about 0.7 kW all night, every night: about 8–9 kWh, close to 40% of daily use. Every 0.3 kW cut saves about 2.6 kWh of battery each night.

6. Inverter runs hot

The inverter's internal temperature was 58 °C at only 2 kW output (18 Oct 2026, 11:37). Inverters reduce output when they get too hot.

7. Earnings figures in the app

SolisCloud values every kWh produced at a flat 4.7 THB and grid power at 0 THB. The earnings shown in the app are not meaningful until import and export prices are set.

8. Room to expand

The 10 kW inverter's second input (MPPT2) is free. A second string of about 4 kWp could add roughly 15 kWh/day on average. The battery already covers the night, so most of that would be exported; it pays only with a sell-back contract.

Recommended actions

Priority	Action	Expected benefit
1	Ask the installer why export was blocked from 28 Mar to 20 Apr, and lock the export setting	Prevents another loss like the 500 kWh in April
2	Confirm the panel count and wattage, and correct the plant capacity in SolisCloud	Accurate performance figures and alerts
3	Apply for (or confirm) a PEA sell-back contract	Paid export instead of free export
4	Check the battery brand setting and the battery-to-inverter communication; recalibrate the charge level	Correct charge readings and longer battery life
5	List what runs overnight and switch off what isn't needed	About 2.6 kWh of battery saved per night for each 0.3 kW cut
6	Check the inverter's ventilation and shading from direct sun	Avoids output cuts on hot days
7	Set the import and export tariffs in SolisCloud	Correct earnings figures in the app

8	Once sell-back is in place, consider a second string of about 4 kWp on MPPT2	Roughly 15 kWh/day more on average

Notes on the data

- Source: SolisCloud daily totals since 29 Jan 2026, and 5-minute inverter data for 1 and 17 Oct 2026 (older 5-minute data is no longer stored).
- The array size of about 6 kWp is estimated from the string's current and voltage. Typical values (about 23 kWh/day for 6 kWp) are approximate yearly averages for southern Thailand, not a site-specific simulation.
- Rev02 replaces Rev01, which assumed two strings and reported the plant as underperforming.
- Prepared by Solar ESS from remote monitoring only; findings 4 and 6 should be confirmed on site.